

Stability Analysis of the CV and CC Control Loops

2-Channel Pre-Regulated Power Supply — study loops (simple_loops.kicad_sch, sheets V_loop and I_loop)

Companion document to V_loop_transfer_functions.ipynb and I_loop_transfer_functions.ipynb — 2026-07-04

1 Overview and method

Both regulator loops share the same four-block structure (Figure 1): an op-amp **error amplifier** with a frequency-shaping feedback network (the compensator), a **driver + pass stage** (small NPN driving a TIP125 PNP Darlington), an **output network**, and a **sense amplifier** that scales the regulated quantity back to the error amp.

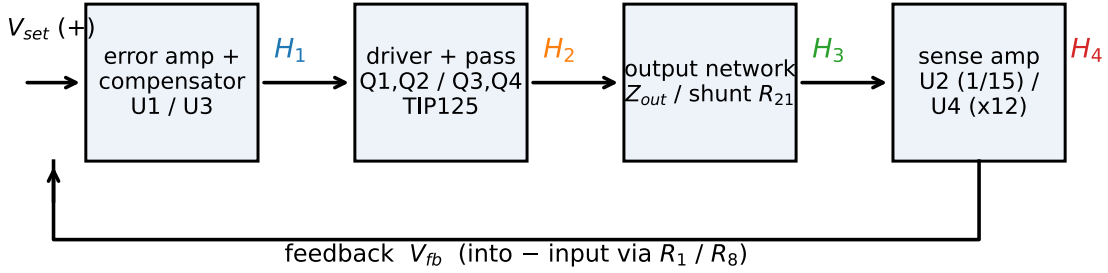


Figure 1: Common loop structure. The voltage loop senses V_{out} (U2, gain 1/15); the current loop senses the shunt voltage $R_{21}I$ (U4, gain 12).

Each block is described by its small-signal transfer function $H_1 \dots H_4$; the **loop gain** is

$$T(s) = H_1(s) H_2(s) H_3(s) H_4(s).$$

The error amp is inverting for the feedback signal, which supplies the 180° that makes the feedback negative; stability is then read from the frequency shaping of T :

- **crossover** f_c : the frequency where $|T| = 1$ (0 dB);
- **phase margin** $\text{PM} = 180^\circ + \angle T(j2\pi f_c)$ — distance from oscillation;
- **gain margin** $\text{GM} = -|T|_{\text{dB}}$ where $\angle T = -180^\circ$;
- **sensitivity peak** $S_{\text{pk}} = \max|1/(1+T)|$ — a single robustness number ($S_{\text{pk}} < 1.5$: well damped; $S_{\text{pk}} > 2$: pronounced ringing).

Because several parameters are uncertain in reality, every design is evaluated over **corners**, not just nominal values: the TIP125 gain spread ($\beta_0 = 500 \dots 4000$), its frequency derating at low bias current, and (for the V loop) the load impedance and the output-capacitor ESR.

2 Device data used in the models

Op-amps (U1–U4): gain-bandwidth product $\text{GBW} = 5.7 \text{ MHz}$, slew rate $2.8 \text{ V}/\mu\text{s}$. The open-loop gain is modelled as a dominant-pole integrator $A(s) = 2\pi \cdot \text{GBW} / s$ and enters the compensator exactly through the noise-gain form (Section 3.1). Slew rate is a large-signal limit and does not appear in the AC model.

TIP125 PNP Darlington (Q2/Q4): the datasheet's small-signal current gain (Fig. 7, PNP @ 3 A) is fitted by a two-pole roll-off,

$$\beta(s) = \frac{\beta_0}{(1 + s/\omega_{\beta 1})(1 + s/\omega_{\beta 2})}, \quad f_{\beta 1} \approx 60 \text{ kHz}, \quad f_{\beta 2} \approx 300 \text{ kHz},$$

which reproduces the curve well (Figure 2). **This roll-off — not the op-amp — is the dominant high-frequency dynamic of both loops.** Two spreads are swept as corners:

- β_0 : datasheet minimum 1000, typical 2000–4000 (DC gain, Fig. 9); 500 is added as cold/low-current pessimism. Loop gain scales 1:1 with β_0 , so the crossover slides almost a decade across production spread.

- Low-bias derating: Fig. 7 is specified at 3 A. At small collector current f_T drops, so a derated corner with $f_{\beta 1,2} = 20/100$ kHz is checked (“LB” in the tables).

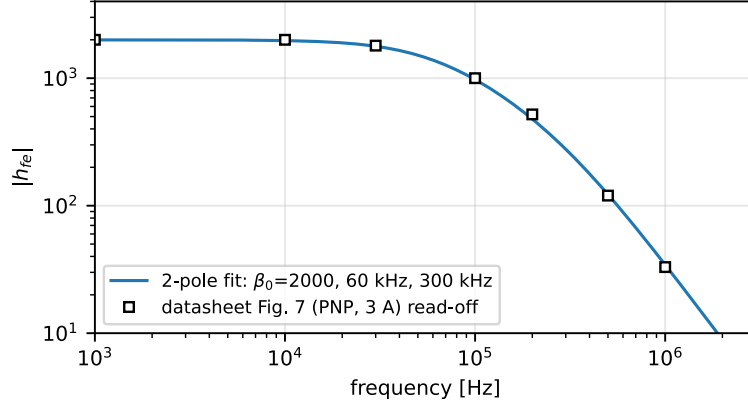


Figure 2: Two-pole fit of the TIP125 small-signal current gain vs. datasheet Fig. 7 read-off points (PNP, $I_C = 3$ A, $V_{CE} = 4$ V).

Diodes D1/D2 (bias steering into the driver base) appear only as the divider $R_3/(R_3 + r_d) \approx 0.97$ with $r_d \approx 26 \Omega$.

3 Voltage loop (sheet V_loop, p. 2)

3.1 Stage 1 – error amplifier / compensator U1

V_{set} enters the (+) input; the feedback voltage V_{fb} enters the (–) input through R_1 , with the feedback impedance Z_f from output to (–):

$$Z_f(s) = \left(R_2 + \frac{1}{sC_1} \right) \parallel \frac{1}{sC_3}, \quad H_1(s) = -\frac{Z_f/R_1}{1 + \frac{1+Z_f/R_1}{A(s)}}.$$

This is a **type-II compensator**: an integrator $1/(sR_1(C_1 + C_3))$, a zero at $f_z = 1/(2\pi R_2 C_1)$ and a high-frequency pole at $f_p = (C_1 + C_3)/(2\pi R_2 C_1 C_3)$. With the optimized values $R_1 = 10$ k, $R_2 = 18$ k, $C_1 = 47$ n, $C_3 = 22$ p:

$$f_z \approx 188 \text{ Hz}, \quad \text{mid-band gain } \frac{R_2}{R_1} = 1.8 \text{ (5.1 dB)}, \quad f_p \approx 400 \text{ kHz}.$$

The finite GBW adds an effective pole near $\text{GBW}/(1 + R_2/R_1) \approx 2$ MHz – a minor extra lag, but included exactly.

3.2 Stage 2 – driver and pass transistor (Q1, R3, R4, D1, Q2)

U1’s output drives Q1 (common emitter, degeneration R_4) through D1; Q1’s collector current is the base current of the TIP125, giving a transconductance

$$H_2(s) = \frac{i_{c2}}{V_e} = \underbrace{\frac{R_3}{R_3 + r_d}}_{\approx 0.97} \cdot \underbrace{\frac{1}{R_4 + r_{e1}}}_{g_{m1}} \cdot \beta_2(s) \text{ [A/V]}, \quad G_m = \frac{0.97 \cdot \beta_0}{150 \Omega} = 13 \text{ A/V (at } \beta_0 = 2000).$$

3.3 Stage 3 – output network (R5, C2, load R6)

Q2’s collector acts as a current source into the output capacitor branch and the load:

$$H_3(s) = Z_{out}(s) = \left(R_5 + \frac{1}{sC_2} \right) \parallel \left[R_{load} \parallel \left(R_{esrL} + \frac{1}{sC_{load}} \right) \right].$$

The load “R6” is **whatever the user connects**, so Z_{out} – and with it the whole loop – changes with the load: a resistive load leaves the $1/(\omega C_2)$ slope up to the ESR zero, while a large load capacitor drags the crossover down by decades. This is the defining difficulty of the voltage loop.

3.4 Stage 4 – sense amplifier U2

A resistive difference amplifier (plus the op-amp’s GBW pole at $\text{GBW}/(1 + R_9/R_{11}) \approx 5.3$ MHz):

$$H_4 = \left(1 + \frac{R_9}{R_{11}}\right) \frac{R_7}{R_7 + R_{10}} \cdot \frac{1}{1 + s/\omega_{U2}} = \frac{1}{15} \cdot \frac{1}{1 + s/\omega_{U2}},$$

so the loop regulates $V_{\text{out}} = 15 V_{\text{set}}$.

3.5 Loop gain and compensator design

Figure 3 shows the assembled loop at the nominal load ($100 \Omega \parallel 1 \mu\text{F}$): the integrator crosses the flat mid-band on a clean single slope, the zero at 188 Hz has long since returned the compensator phase, and the crossover lands at $f_c \approx 22 \text{ kHz}$ with $\text{PM} \approx 76^\circ$.

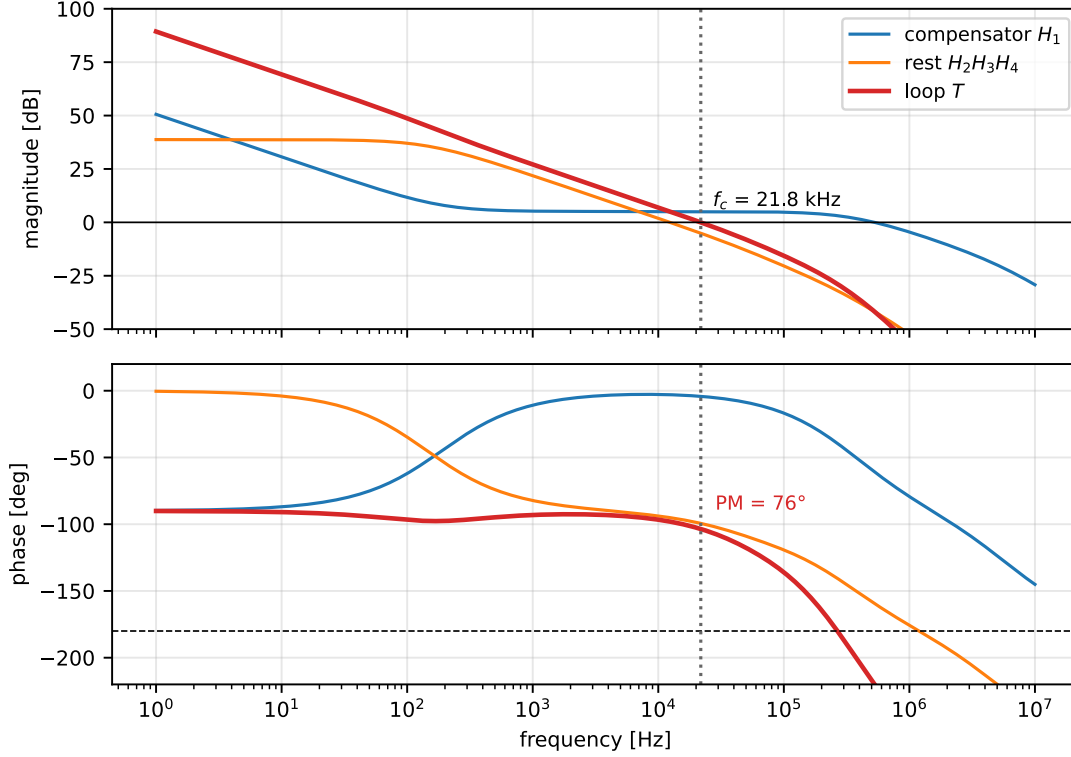


Figure 3: V loop at $\beta_0 = 2000$, load $100 \Omega \parallel 1 \mu\text{F}$: compensator H_1 , rest $H_2H_3H_4$, and loop gain T .

The values were found by a grid search over E12 components with the corner requirements $\text{PM} \geq 45^\circ$ ($\geq 35^\circ$ for the extreme $1000 \mu\text{F}$ case), $\text{GM} \geq 10 \text{ dB}$ and $S_{\text{pk}} \leq 1.7$ over **all** corners (β spread $\times$ ESR $\times$ load set), maximising the worst-case crossover. Two rules of thumb summarise why the result looks the way it does:

1. **The zero must sit below the biggest-cap crossover.** A large load capacitor drags f_c down; if f_c falls below f_z the loop reverts to integrator-on-top-of-cap-pole (-180° slope) and the margin collapses. $f_z = 188 \text{ Hz}$ keeps even the $\beta = 500 + 1000 \mu\text{F}$ corner ($f_c \approx 116 \text{ Hz}$) inside the zero's phase lead.
2. **The mid-band gain sets the light-load crossover.** $R_2/R_1 = 1.8$ keeps the crossover at $\beta_0 = 4000$ ($f_c \approx 45 \text{ kHz}$) clear of the β poles even when the design is denied the help of C2's ESR zero (ESR corner $50 \text{ m}\Omega$).

3.6 Robustness across loads and corners

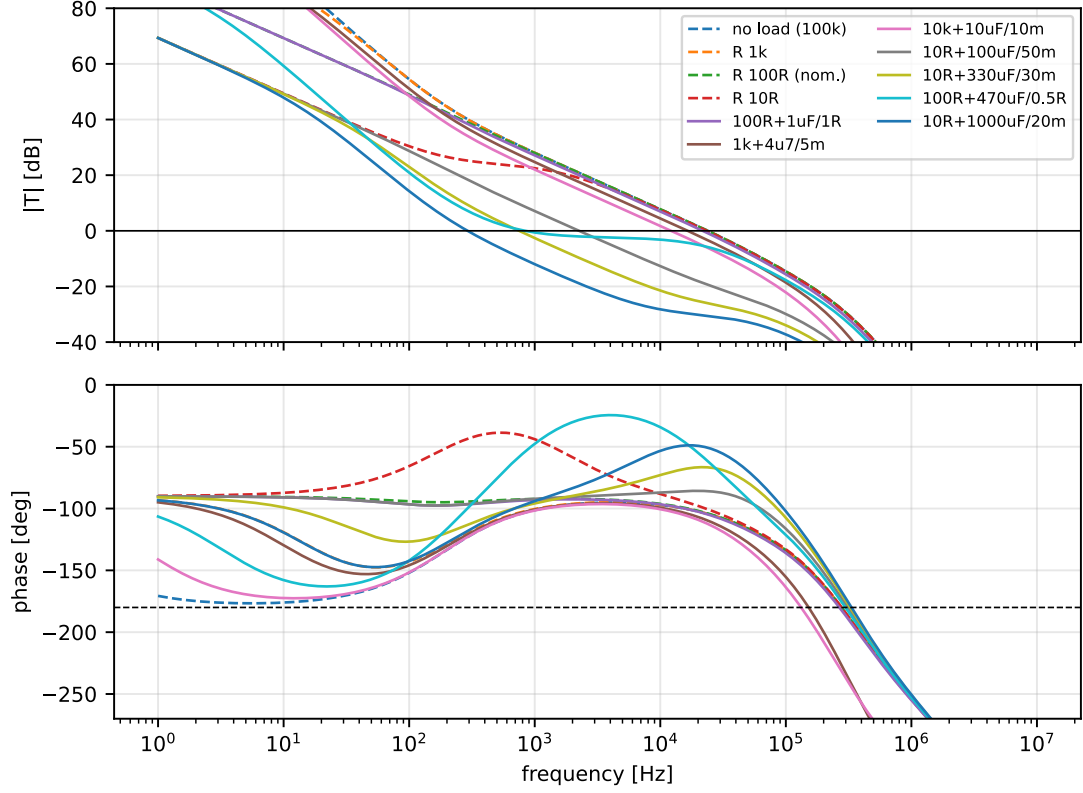


Figure 4: V-loop gain across 11 load types at $\beta_0 = 2000$ (dashed: purely resistive). Big capacitors move the crossover down by up to two decades — the compensator must be stable along the whole trajectory.

load case	f_c [Hz]	PM [°]	GM [dB]	S_{pk}	worst PM [°]	worst S_{pk}
no load (100 k)	23 930	75.6	28.1	1.15	37.0	1.99
1 k Ω	23 926	75.7	28.1	1.15	37.0	1.99
100 Ω (nominal)	23 885	76.1	28.1	1.15	37.5	1.97
10 Ω (heavy)	23 440	79.7	28.3	1.14	42.2	1.83
100 Ω + 1 μ F/1 Ω	21 801	76.4	28.9	1.15	40.2	1.87
1 k + 4.7 μ F/5 m Ω	16 534	75.8	24.3	1.17	46.1	1.69
10 k + 10 μ F/10 m Ω	12 240	77.6	26.3	1.15	53.6	1.51
10 Ω + 100 μ F/50 m Ω	2 252	89.7	44.2	1.01	84.5	1.05
10 Ω + 330 μ F/30 m Ω	749	81.1	48.7	1.01	62.7	1.03
100 Ω + 470 μ F/0.5 Ω	838	126.1	32.2	1.07	61.0	1.26
10 Ω + 1000 μ F/20 m Ω	292	61.9	52.2	1.01	40.0	1.48

Table 1: V loop per load: left half at the typical corner ($\beta_0 = 2000$, $f_\beta = 60/300$ kHz, ESR 0.2 Ω); right half the worst case over all β /ESR/low-bias corners.

β corner	f_c range [Hz]	worst PM [°]	min GM [dB]	max S_{pk}
$\beta = 500$	116 ... 6 188	40.0	35.0	1.48
$\beta = 1000$ (min spec)	178 ... 12 279	49.5	28.9	1.21
$\beta = 2000$ (typical)	292 ... 23 930	61.9	22.9	1.33
$\beta = 4000$ (typ. peak)	522 ... 44 766	47.8	16.9	1.64
$\beta = 1000$, low-bias	178 ... 10 920	49.1	19.6	1.49
$\beta = 2000$, low-bias	291 ... 18 403	37.0	13.6	1.99

Table 2: V loop worst case per β corner, over all 11 loads and both ESR values. The binding corners are $\beta = 4000$ with low-ESR C2 (fast side) and the doubly derated low-bias case (slow side); both remain stable with sensible margin.

3.7 Setpoint step and prefilter

The closed-loop response from the setpoint is

$$\frac{V_{out}}{V_{set}} = \frac{G_m(s) Z_{out} (1 + Z_f/R_1)}{1 + T},$$

whose numerator carries the **noise-gain** term $(1 + Z_f/R_1)$ – the compensator zero appears in the reference path with a zero at

$$f_{z,ref} = \frac{1}{2\pi(R_1 + R_2)C_1} \approx 121 \text{ Hz.}$$

A setpoint edge therefore feeds forward and overshoots by $\approx 54\%$ – **not** ringing (the loop is critically damped); it disappears if the setpoint never contains energy above that zero. The clean fix is an RC low-pass in front of the (+) input with its pole **matched to the zero**: $\tau = (R_1 + R_2)C_1 \approx 1.32 \text{ ms}$, e.g. **27 k + 47 n**. The zero then cancels the filter pole (Figure 5). Note that $R_1 C_1$ alone ($470 \mu\text{s}$) is not the match – it leaves about 13 %.

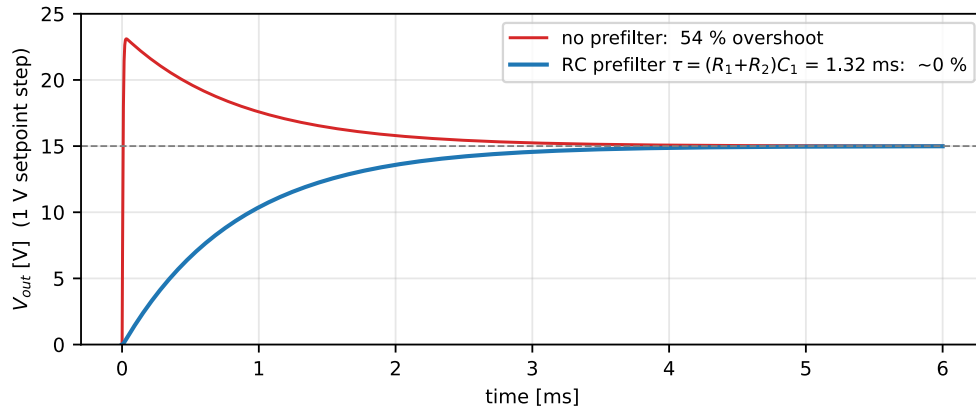


Figure 5: Closed-loop 1 V setpoint step ($\beta_0 = 2000$, nominal load) with and without the matched RC prefilter.

4 Current loop (sheet I_loop, p. 3)

4.1 Stages 1, 2 and 4

The error amplifier U3 and the driver Q3/Q4 are identical in topology to the voltage loop ($R_8 = 10 \text{ k}$ input resistor, $R_{15} = 150 \Omega$ degeneration, same TIP125 model). The optimized compensator is a **pure integrator** – R_{12} and C_5 stay unpopulated:

$$H_1(s) = -\frac{1/(sR_8C_4)}{1 + (1 + 1/(sR_8C_4))/A(s)}, \quad C_4 = 27 \text{ n} \rightarrow \text{unity gain at } \frac{1}{2\pi R_8 C_4} \approx 589 \text{ Hz.}$$

The sense amplifier U4 is non-inverting with gain $1 + R_{16}/R_{17} = 12$; its noise gain of 12 pulls the op-amp pole down to $\text{GBW}/12 \approx 475 \text{ kHz}$ – low enough to matter alongside the β poles:

$$H_4(s) = \frac{12}{1 + s/(2\pi \cdot 475 \text{ kHz})}, \quad V_{fb} = 12 \cdot R_{21} \cdot I = 2.4 \text{ V/A} \quad (I \approx 417 \text{ mA per volt of } V_{set}).$$

4.2 Stage 3 – the flat, load-independent plant

Here the schematic differs decisively from the voltage loop: the output-capacitor branch ($R_{19} + C_6$) **and** the load R_{20} both return to the **top of the shunt** R_{21} , not to ground. Every milliampere that Q4 sources must therefore pass through the shunt, and Kirchhoff's current law gives, exactly and at every frequency,

$$i_{R_{21}} = i_{C_6} + i_{\text{load}} = i_{c4} \Rightarrow H_3(s) = \frac{V_{\text{sh}}}{i_{c4}} = R_{21} = 0.2 \, \Omega.$$

The load impedance, C_6 and R_{19} **drop out of the loop gain completely** – they only set the voltage at the output node, which the regulated variable never sees. The loop is identical for a short, a resistor, or 1000 μF . Including Q4's finite output resistance ($r_o \approx 2 \text{ k}$ pessimistic) and output capacitance ($C_{\text{ob}} \leq 300 \text{ pF}$) merely adds a static gain error up to $\approx 5 \%$ at light load (a current-divider effect, calibration territory) and **no dynamics** below the MHz range.

4.3 Loop gain – why a pure integrator

With a flat plant, the “rest” of the loop is a constant $0.97\beta_0 R_{21} \cdot 12/R_{15} \approx 31$ (30 dB at $\beta_0 = 2000$) with the $\beta/\text{U4}$ poles hanging off the top (Figure 6). A single integrator crosses it on a clean -20 dB/dec slope: crossover

$$f_c \approx \frac{0.97 \beta_0 R_{21} (1 + R_{16}/R_{17})}{2\pi R_8 C_4 R_{15}} = 17.6 \text{ kHz at } \beta_0 = 2000.$$

A voltage-loop-style zero ($R_{12} > 0$) would be actively harmful here: it flattens the compensator above f_z , and with the 8:1 β -gain spread the crossover would slide across that flat region straight into the β poles. **Keep $R_{12} = 0$.** $C_4 = 27 \text{ n}$ is the speed/margin knee: 22 n is $\approx 20 \%$ faster but sits at the acceptance edge (low-bias $S_{\text{pk}} \approx 2$), 47 n is the conservative fallback ($\text{PM} \geq 65^\circ$).

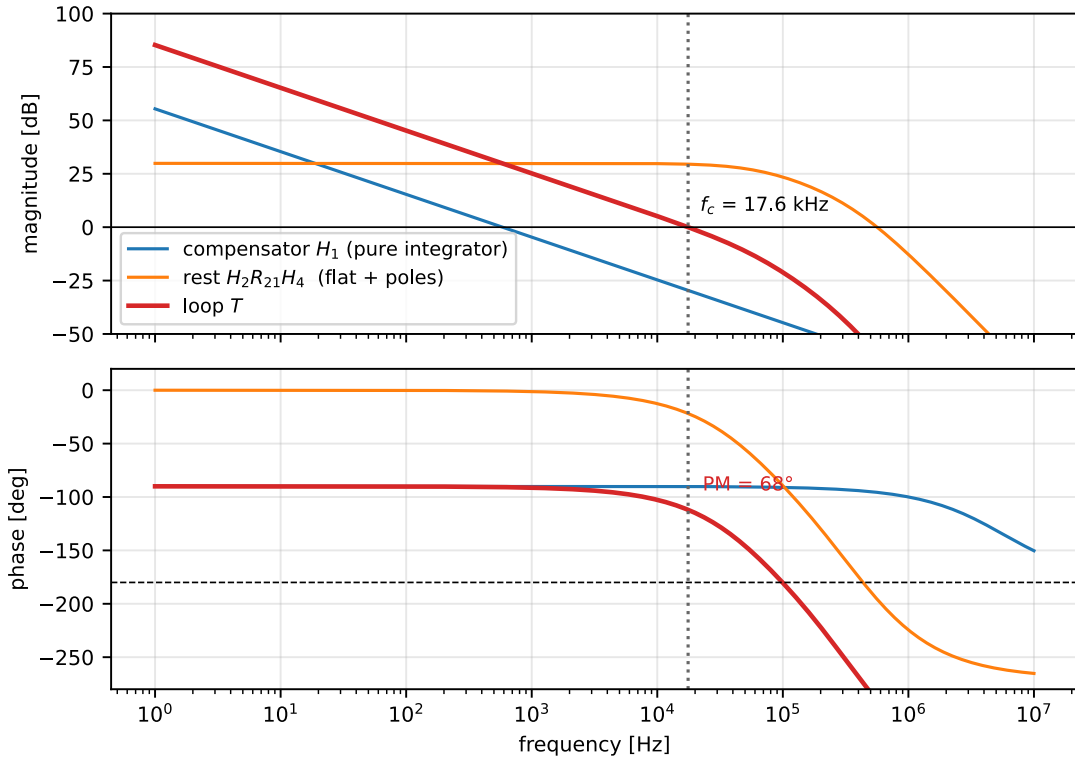


Figure 6: I loop at $\beta_0 = 2000$: integrator, flat rest, and loop gain. The phase budget at crossover is -90° (integrator) minus the β -pole lag.

4.4 Corner sweep

β corner	f_c [Hz]	PM [°]	GM [dB]	S_{pk}
$\beta = 500$	4 582	84.2	33.1	1.08
$\beta = 1000$ (min spec)	9 082	78.5	27.1	1.15
$\beta = 2000$ (typical)	17 597	68.0	21.0	1.30
$\beta = 4000$ (typ. peak)	32 148	51.5	15.0	1.60
$\beta = 1000$, low-bias	8 437	61.2	20.3	1.39
$\beta = 2000$, low-bias	14 662	43.5	14.3	1.78

Table 3: I loop margins per β corner. The numbers are **identical for every load** – the load does not appear in $T(s)$ (Section 4.2).

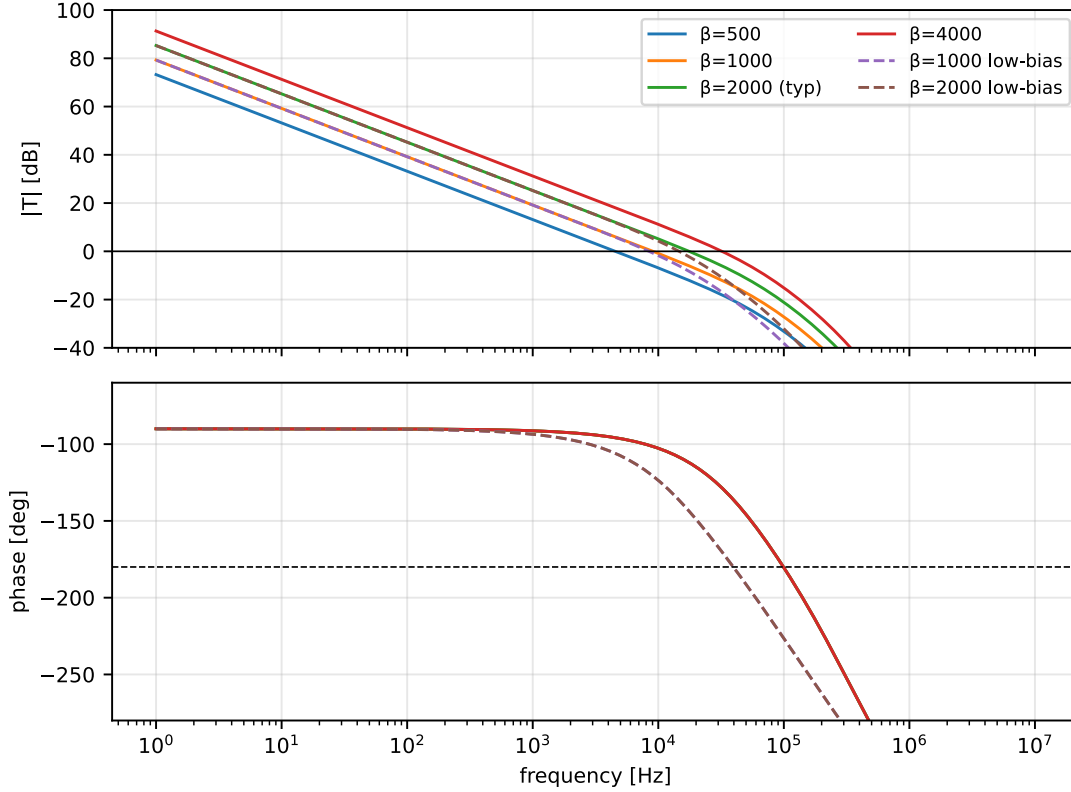


Figure 7: I-loop gain across the TIP125 β corners (dashed: low-bias derated poles). Only the gain scales; the pole structure – and therefore the phase curve – barely moves.

4.5 Setpoint step – the feed-forward spike

Because the plant has **no inertia** (a flat 0.2Ω , no output pole in the current path), a V_{set} edge feeds through the noise-gain term $(1 + Z_f/R_8) \rightarrow 1$ directly to the driver and commands $\Delta I \approx G_m \Delta V_{set} \approx 13 \text{ A/V}$ for the few microseconds ($1/2\pi f_c \approx 9 \mu\text{s}$) the integrator needs to take over – the small-signal model shows a spike of more than 2000 % (Figure 8, left). Hardware clips it (op-amp swing and $2.8 \text{ V}/\mu\text{s}$ slew, Q3's emitter resistor), but the lesson stands: **the current setpoint must never step hard**. The matched RC prefilter (pole on the reference zero at $1/2\pi R_8 C_4 = 589 \text{ Hz}$, i.e. $\tau = R_8 C_4 = 270 \mu\text{s}$, **10 k + 27 n**) removes the spike while keeping the setpoint response fast, because the compensator zero cancels the filter pole (Figure 8, right).

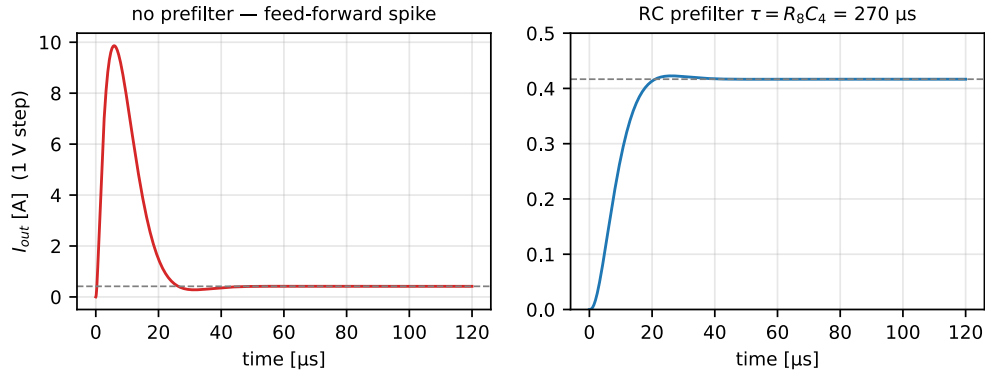


Figure 8: I loop, 1 V setpoint step at $\beta_0 = 2000$. Left: unfiltered — the feed-forward spike reaches amperes for a few μs (model; hardware clips it). Right: with the matched 270 μs prefilter — $\approx 1.4\%$ residual, settled in $\approx 20\ \mu\text{s}$.

5 Setpoint prefilters — general rule

Both overshoots have the same root cause: the compensator's feedback impedance appears in the **reference path** as $(1 + Z_f/R_{\text{in}})$, whose zero lets a setpoint edge feed forward. The matched RC prefilter puts its pole exactly on that zero:

$$\tau_{\text{filt}} = (R_{\text{in}} + R_{\text{zero}}) \cdot C_{\text{int}}$$

loop	reference zero	matched τ	suggested RC	result (1 V step)
V loop	$1/2\pi(R_1 + R_2)C_1 \approx 121\ \text{Hz}$	1.32 ms	27 k + 47 n (or 10 k + 120 n)	54 % $\rightarrow \approx 0\%$
I loop	$1/2\pi R_8 C_4 \approx 589\ \text{Hz}$	270 μs	10 k + 27 n	2267 % $\rightarrow 1.4\%$, settle $\approx 20\ \mu\text{s}$

Table 4: Matched setpoint prefilters. The filter is a series R + C-to-ground directly at the error amp's (+) input (high impedance). A firmware DAC ramp of $\approx 1\ \text{ms}$ is the part-free alternative. Note the values track the compensator: change C_1/C_4 and the filter caps follow.

Because the reference zero contains only compensator parts, the cancellation is independent of the TIP125 β and of the load.

6 Summary

part	schematic first guess	final value	role
V: R_1	10 k	10 k (keep)	input / gain reference
V: R_2	10 k	18 k	zero at 188 Hz, mid-band gain 1.8
V: C_1	1 n	47 n	integrator; zero with R_2
V: C_3	—	22 p	HF pole $\approx 400\ \text{kHz}$
I: R_8	10 k	10 k (keep)	input / gain reference
I: C_4	—	27 n	pure integrator, unity at 589 Hz
I: R_{12}, C_5	—	DNP	a zero would be harmful (flat plant)
V prefilter	—	27 k + 47 n	kills 54 % setpoint overshoot
I prefilter	—	10 k + 27 n	kills the feed-forward current spike

Table 5: Final component values (all E12).

Resulting worst-case margins — V loop: PM $\geq 48^\circ$, GM $\geq 17\ \text{dB}$ over all loads (short of the doubly derated low-bias corner, which bottoms at 37°); typical- β margins $62\text{--}126^\circ$. I loop: PM $\geq 51.5^\circ$, GM $\geq 15\ \text{dB}$ over the full β spread (low-bias corner 43.5°), for **any** load.

Not covered by these AC models, to be verified on hardware: the CV $\leftrightarrow$ CC handover through the D1/D2 diode-OR (the losing loop's integrator saturates and must slew back — an anti-windup clamp shortens the transition), the β -pole derating assumed for light loads (a small bleeder keeping tens of mA through the pass device firms it up), and thermal drift of the TIP125 parameters.